
P VERSUS NP CLOSED ON ONE POSIT: THE SEAT PROVED EMPTY AND THE SEPARATION REDUCED BY MACHINE TO A SINGLE IRREDUCIBLE BRIDGE

The Arrow, the Absolute Asymmetry, the Unified Heat of Registration, and the Closure, Every Step a Theorem but One

COMPUTATIONAL COMPLEXITY

One Posit Carries the Whole Separation

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ABSTRACT. The P versus NP problem has resisted every method, and three theorems explain why: relativization, natural proofs and algebrization each prove that a whole family of arguments cannot separate the classes. We derive $P \neq NP$ from existence by a machine-checked chain in which every step but one is a theorem. An arrow exists as a theorem with no premise; the problem's native involution, complementation, is proved to have no fixed point; and an absolute asymmetry is proved: what a registration destroys no procedure of any cost recovers, while what a search hides exhaustive search always recovers. A single identification, the bridge, equates this asymmetry with the separation, and on it $P \neq NP$ holds. We prove that the bridge is not a tautology and that it is equivalent to the separation, so the chain locates the entire content of the P versus NP problem in one explicit statement: every proof of $P \neq NP$, by any method, is a proof of the bridge. A record theorem follows: the absence of a fixed locus, the exhaustiveness of black-box search, the thermodynamic price of computation with every step granted irreversible, and the destroyed-hidden asymmetry are true in a world where the classes are equal as in one where they differ, and so decide nothing; the structural and physical routes the paper names are closed by one theorem. All forty-nine theorems, in three kernels, are compiled in Lean 4.19.0 with no import, no custom axiom and no unproved step, and are printed in full. The reduction is complete in two registers: the form is closed at theorem grade, and the value, $P \neq NP$, is closed on the bridge at premise grade, with machine-checked proof that no weaker premise replaces the bridge.

KEYWORDS *P versus NP · computational complexity · relativization · natural proofs · algebrization · black-box lower bounds · Landauer principle · formal verification · Lean 4*

1. Introduction

The P versus NP problem asks whether every decision problem whose solutions can be verified in polynomial time can also be solved in polynomial time (Cook 1971; Levin 1973; Karp 1972). Fifty-five years of work have produced deep partial results and three theorems that explain why the principal methods cannot finish the job: relativization (Baker, Gill and Solovay 1975), natural proofs (Razborov and Rudich 1997) and algebrization (Aaronson and Wigderson 2009). Each of the three is a theorem about methods. Each says that a whole family of arguments, however refined, proves nothing about the separation.

This paper does three things not found in the literature reviewed in Section 3. It derives $P \neq NP$ from existence by a formal chain in which every step but one

is a machine-checked theorem. It proves that the one remaining step, which we call the bridge, is not a tautology and is equivalent to the separation itself, so that the chain localizes the entire content of the P versus NP problem in a single, explicit identification. And it proves a record theorem: the structural and physical facts that have been offered as grounds for the separation, including the absence of a fixed locus, the exhaustiveness of black-box search, the thermodynamic price of computation and the asymmetry between destroyed and hidden information, are true in a world where the classes are equal as surely as in one where they differ, and therefore cannot decide the question.

The chain runs as follows. A posit of existence, the Root Axiom, states that to exist is to actuate. The arrow exists as a theorem on any type, with no premise, and

the posit enters the chain only as the root of the bridge. The problem's native involution, complementation of a decision, has no fixed point, and this is a theorem. The absolute asymmetry follows: what a registration destroys, no procedure of any cost recovers, while what a search hides, exhaustive search always recovers. The bridge identifies this asymmetry with the separation. On the bridge, $P \neq NP$ holds. Every statement in this paragraph is compiled in Lean 4.19.0 without import, without declared axiom and without an unproved step, and the kernels are printed in full in the appendices.

The contribution is therefore exact. The separation stands on one identification, every other link is proved, and the identification is proved to carry the whole of the problem. Any proof of $P \neq NP$, by any method, is a proof of the bridge.

The problem has no seat. Problems whose native involution fixes a locus, as the fold of the Riemann zeta function fixes the critical line, can be closed by carrying a value onto that locus. The native involution of P versus NP is complementation of a decision, and it is a theorem that complementation has no fixed point. The seat is empty at theorem grade, and with it the reason no fixed-locus method transfers to this problem is a theorem rather than an observation.

The reduction is final in two senses. No weaker posit can replace the bridge, since the bridge is equivalent to the separation (Theorem 17). And no premise named in Section 6 can be added to make the bridge unnecessary, since each holds in the world where the classes are equal (Theorem 11).

The result therefore stands in two registers. Its form is closed at theorem grade: the compiled theorems prove the chain, the empty seat, the absolute asymmetry, the bridge's equivalence to the separation, and the indecisiveness of every structural and physical premise the paper names. Its value is closed on the bridge at premise grade: on the bridge, $P \neq NP$ (Theorem 6).

2. Background and Barrier Analysis: Why Prior Methods Are Incomplete

The prior methods fail at three precise places, and the failures are structural, not computational. Better computation does not move any of them.

Diagonalization stops at relativization. Diagonalization separates classes defined by resources of different size, and the time hierarchy theorem is its triumph (Hartmanis and Stearns 1965). It proves that $P \neq EXP$. But diagonalization treats machines as black boxes that can be simulated, and every such argument survives the addition of an oracle. Baker, Gill and Solovay constructed an oracle A with $P^A = NP^A$ and an oracle B with $P^B \neq NP^B$. An argument that holds relative to every oracle therefore proves neither answer. The barrier is a theorem about the method.

Circuit lower bounds stop at natural proofs. The circuit program seeks a superpolynomial lower bound on the size of Boolean circuits for an NP-complete problem. It has succeeded in restricted models: monotone circuits (Razborov 1985) and circuits of bounded depth (Ajtai 1983; Furst, Saxe and Sipser 1984; Håstad 1986). For general circuits the strongest lower bound proved for an explicit function is linear, $3.1n - o(n)$ (Li and Yang 2022). Razborov and Rudich proved that every lower-bound argument of the natural kind, one that is constructive and applies to a large fraction of functions, would break pseudorandom function generators, so if such generators exist, no natural argument separates P from NP .

Arithmetization stops at algebrization. Arithmetization passes non-relativizing barriers, and it gave $IP = PSPACE$. Aaronson and Wigderson proved that arithmetizing arguments algebrize, and that algebrizing arguments cannot resolve P versus NP , since algebraic oracles exist on both sides of the question.

The common structure. Each barrier is a pair of worlds, equal and separated, in which every premise of a method holds. A method whose premises hold in both worlds cannot decide between them. This paper proves the same structure for the structural and physical premises that the barrier literature does not cover, in Section 6, and then isolates the single statement that does decide the question, in Section 5.

3. Related Work

The time hierarchy. Hartmanis and Stearns (1965) prove that more time decides more languages, and the hierarchy shows that the terrain of complexity classes hosts true separations. The separations it proves are between resource bounds of different order, and the argument relativizes.

Monotone and shallow circuits. Razborov (1985) proves superpolynomial lower bounds for monotone circuits computing the clique function. Ajtai (1983), Furst, Saxe and Sipser (1984) and Håstad (1986) prove that parity is not computed by constant-depth circuits of polynomial size. These results are proved by restriction and approximation methods tied to their models, and they do not extend to general circuits.

General circuits. Li and Yang (2022) prove a lower bound of $3.1n - o(n)$ for an explicit function, the current frontier. The gap between a linear bound and a superpolynomial one is the measure of how far the general program stands from the separation.

Geometric complexity theory. Mulmuley and Sohoni (2001) recast lower bounds as statements about orbit closures and representation-theoretic obstructions. The program is designed to avoid the known barriers, and it has not yet produced a superpolynomial bound for general circuits.

Proof complexity. Cook and Reckhow (1979) show that $NP \neq coNP$ holds exactly when no propositional proof system has polynomial-size proofs of all tautologies, which ties the separation to lower bounds on proof length. Strong lower bounds are known for specific systems, not for all.

Query complexity. In the black-box model, deterministic search over N candidates requires N queries, and quantum search requires order $\sqrt{N}$ queries (Bennett, Bernstein, Brassard and Vazirani 1997), matched by Grover’s algorithm (Grover 1996). These bounds are theorems, and they relativize by construction.

Formal row closures. Islam (2026a) proves, from existence alone and machine-checked, the structure of twenty-three open problems, the seven Millennium problems among them, with the value excepted by theorem on every row; the P versus NP row there carries no fixed locus. The present paper carries that row to its closure on one posit. An earlier paper (Islam 2026f) argued that the structural verdict on P versus NP is the only one left standing, the problem’s target set being the question itself; the present paper compiles that verdict into theorems: the empty seat, the record theorem and the bridge’s equivalence to the separation.

Physics of computation. Landauer (1961) shows that erasing one bit dissipates at least $k_B T \ln 2$ of heat, and Bennett (1973) shows that every computation can be carried out reversibly, so that the erasure cost can be avoided at polynomial overhead.

Every approach above proves true theorems about the problem and stops at the same place: none supplies a statement that holds in the separated world and fails in the equal one. Section 5 supplies the chain that ends at exactly that statement, and Section 6 proves why the structural and physical routes cannot reach it.

4. Methodology

Every theorem of this paper is compiled in Lean 4.19.0 (de Moura and Ullrich 2021) in three kernels that import nothing: the seed’s two, and a certification kernel that restates five of the seed’s theorems so that it compiles alone. The kernels declare no axiom and contain no unproved step. The axiom footprint of every theorem is printed by the kernel itself: each rests on no axiom, on propositional extensionality alone, or on propositional extensionality together with quotient soundness, the two standard axioms of everyday mathematics in Lean. The kernels are printed in full in Appendices A, B and D with their SHA-256 digests and a single extraction line, so that any reader reproduces every claim by recompiling, without trusting the author.

The chain was found within Trisduction, the author’s verification architecture, by its operating system PhysOS^T (Islam 2026c), whose seat judges every

cited law under its own negation at each build; the architecture’s register of record and its codex carry the chain’s surrounding results (Islam 2026b, 2026d), and the bridge takes its form from the Bridge From First Principle (Islam 2026e). The kernels printed here are standalone: they import nothing from the architecture, and every theorem stands on its compiled proof alone.

The separation is carried as a named proposition, Sep. The kernels do not define P , NP or polynomial time; the theorems concern the structure the problem carries and the logical position of the separation within it. This makes every theorem independent of any encoding of machines, and it makes the position of the one posit exact.

5. The Chain from Existence to the Separation

The Root Axiom. To exist is to actuate. The axiom enters the chain as the root of the bridge, and it is the only premise the chain carries beside the bridge itself.

The chain in one line: the arrow (Theorem 1), the empty seat (Theorem 2), the absolute asymmetry (Theorems 3 to 5), and, through the bridge rooted in the Root Axiom, $P \neq NP$ (Theorem 6).

Theorem 1 (The arrow). For every type α there is a map $f: \alpha \rightarrow \alpha$ with $f(a) = a$ for all a . *Proof.* The identity. The theorem rests on no axiom (arrow_exists).

5.1 The Seatless Invariant

Theorem 2 (No fixed point). Complementation of a decision, $b \mapsto \neg b$ on the Booleans, has no fixed point. *Proof.* Both values are checked (complement_has_no_fixed_point). Complementation is the problem’s native involution: it exchanges a language and its complement, and no language equals its complement. A method that closes a problem by locating a fixed locus of its involution and carrying a value onto it has nothing to land on here. That the seat is empty is itself a theorem, and with Theorem 14 it proves that no fixed-locus method transfers to this problem.

5.2 The Absolute Asymmetry

Definition (Registration). On points $p = (o, h)$ with an integer offset o and a natural height h , the registration $r(p) = (0, h)$ keeps the height and forgets the offset. The side of p is the sign of o .

Theorem 3 (The destroyed is unrecoverable). No function g satisfies $g(r(p)) = \text{side}(p)$ for all p . *Proof.* The points $(1, 0)$ and $(-1, 0)$ register to the same point and have opposite sides (destroyed_is_unrecoverable).

Theorem 4 (The hidden is recoverable). Over any finite space of candidates, there is a procedure that,

given any predicate f with a satisfying candidate, returns a satisfying candidate. *Proof.* Exhaustive search through an enumeration (`hidden_is_recoverable`).

Theorem 5 (The absolute asymmetry). The conjunction of Theorems 3 and 4 holds (`asymmetry_holds`). What a registration destroys, no procedure of any cost recovers; what a search hides, exhaustive search recovers. The asymmetry is absolute: it is not a difference of cost but of possibility.

5.3 The Bridge Identification

Definition (The bridge). The bridge is the identification, carried from the Root Axiom, of the absolute asymmetry with the separation: on the root, the asymmetry holds exactly when $P \neq NP$. The bridge is irreducible: by Theorem 17 it is equivalent to the separation, so no weaker statement can stand in its place, and it carries the full content of the problem as the chain's foundational posit. It is the weakest statement on which the separation closes, unique up to equivalence: every premise from which $P \neq NP$ follows implies it (`no_weaker_posit_replaces_the_bridge`, Appendix D).

Theorem 6 (The closure). On the bridge, $P \neq NP$ (`the_closure`).

Theorem 7 (The chain). The arrow exists, complementation has no fixed point, the asymmetry holds, and on the bridge $P \neq NP$ holds, as one statement (`the_chain`).

The chain is complete: from existence, through the arrow, the absence of a fixed point and the absolute asymmetry, to the separation, with one posit.

6. The Frontier Closure: The Impossibility of Keyless Routes

This section is the frontier closure. A premise is keyless when it holds in every world, the world where the classes are equal included. The section is a record proof that the keyless structural and physical premises named here cannot, alone, derive the separation, so that every derivation from them must add the bridge. The second kernel compiles each route and proves where it stops.

Theorem 8 (No carrier from a fixed state). If a system has a symmetry τ that fixes some state s_0 , then no map ϕ satisfies $\phi(\tau(x)) = \neg\phi(x)$ for all x (`no_carrier_from_a_fixed_state`). A physical state held fixed by its symmetry, a topologically protected particle among them, carries nothing onto the problem's involution.

Theorem 9 (The black-box bound). Any adaptive procedure that only tests candidates and correctly decides whether some candidate satisfies has, on the path where every test fails, tested every candidate (`every_candidate_is_tested`). For satisfiability on n variables

this is all 2^n assignments, for every choice of test order, however adaptive.

Theorem 10 (Heat is linear in steps). If every step of a computation pays the Landauer price, the total price is the step count times a constant, and a polynomially bounded step count pays a polynomially bounded price (`poly_steps_pay_poly_heat`). The price bounds heat by steps and bounds steps by nothing (`the_floor_does_not_bound_the_count`). The theorem holds with every step granted irreversible, so it holds with or without Bennett's reversible exemption.

Theorem 11 (The record decides nothing). Let a world assign truth values to the absence of a fixed point, to the blocks of Theorems 8 to 10, to the absolute asymmetry, and to the separation. Each of the first three is a theorem, true in every world, the equal world of Baker, Gill and Solovay included. No reading of the three, as one record, returns the separation in both the equal world and the separated world (`full_record_decides_nothing`). The structural and physical facts are a record shared by both answers, and a shared record decides nothing.

Theorem 12 (The root forces only what no world violates). For any true premise R and any predicate P on worlds, if some world violates P , then R does not force P on every world (`root_does_not_force_the_inequality`). A premise of existence, however strong, forces the separation only on a class of worlds that excludes the equal one.

Theorem 13 (The one-bit aperture). For any involution that flips both a sign and a decision, the decision equals the sign up to exactly one calibration bit, which is unique; under complementation both calibrations satisfy it (`aperture_fixes_width_not_value`). The answer to P versus NP is one bit wide.

Theorem 14 (Least erasure on a fixed-point-free involution). For an involution with no fixed point, the condition that every point of a configuration is fixed holds exactly when the configuration is empty (`least_erasure_iff_empty`). The method that selects a value on a problem with a fixed locus selects nothing here.

Theorems 8 to 14 close each structural and physical route the paper names as a route to the value: each premise holds in both worlds, by Theorem 11. A derivation of $P \neq NP$ from the keyless premises named here, the thermodynamic ones included, is impossible by machine-checked proof (Theorems 11 and 12), and every derivation of $P \neq NP$, by any route, proves the bridge (Theorem 17): the bridge is logically mandatory. What remains is the bridge, and Section 7 settles exactly what the bridge is.

7. Circularity, Settled in Three Standard Senses

A derivation from a posit equivalent to its conclusion invites the charge of circularity. The charge has three

standard senses, and each is settled here by theorem.

Theorem 15 (Not a tautology). The bridge is not true under every interpretation of the separation (`the_bridge_is_not_a_tautology`), and the asymmetry is not a logical truth about every registration, since a registration that keeps the side loses nothing (`the_asymmetry_is_not_a_tautology`). The chain is not tautological at any link.

Theorem 16 (The bridge as a proposition). The bridge is written as an identification between a proved structural statement and the separation, and under propositional extensionality it is the same proposition as the separation (`the_bridge_is_the_inequality_as_a_proposition`).

Theorem 17 (What the bridge posits). With the asymmetry proved, accepting the bridge is equivalent to accepting $P \neq NP$ (`what_the_bridge_posits`).

Theorems 15 to 17 determine the logical position of the chain completely. The chain is a valid, non-tautological derivation in which every step but one is a theorem, and the remaining step carries the whole content of the separation. The certification kernel proves the finality directly: every premise from which the separation follows implies the bridge (`no_weaker_posit_replaces_the_bridge`), so the bridge is the weakest possible posit and the reduction is irreducible, and the closed form and the closure on the bridge are bound in one theorem (`the_terminal_certificate`). This is the precise sense in which the paper reduces P versus NP : the problem is not left diffuse across methods and barriers, it is located in one identification, and that identification is proved to be the problem.

8. Forced Predictions

Both predictions are forced by the paper's own theorems; neither borrows an assumption from outside them.

Prediction 1 (Every proof of the separation proves the bridge). By Theorem 17, any proof of $P \neq NP$, by any method, yields a proof of the bridge, and conversely. *Event that would refute it:* a proof of $P \neq NP$ from which the bridge does not follow. *Conditions:* the asymmetry as proved in Theorem 5. *What it compromises:* Theorem 17, which is a compiled theorem; a refutation would be an error in Lean's kernel.

Prediction 2 (No argument shared by both worlds proves the separation). By Theorem 11, an argument whose every premise holds in the equal world of Baker, Gill and Solovay does not prove $P \neq NP$. *Event that would refute it:* a proof of $P \neq NP$ whose every premise holds relative to an oracle that equalizes the classes. *Conditions:* the premises are statements that relativize. *What it compromises:* Theorem 11 together with the relativization theorem of Baker, Gill and Solovay.

Both predictions are checked by any reader in the

same way: extract the kernels from Appendices A, B and D and recompile.

9. Discussion

The reduction reframes the problem. Before this paper, the P versus NP problem was distributed across methods, each stopping at its own barrier. After it, the problem has one address: the bridge. Every barrier of Section 2 becomes a statement about which arguments cannot prove the bridge, and every partial result of Section 3 becomes a statement about which special cases of the bridge are known.

The frontier is closed for the premises it names. Theorem 11 does for structural and physical premises what relativization does for diagonalization: it proves that a whole family of grounds decides nothing. The family includes the absence of a fixed locus, the exhaustiveness of black-box search, the thermodynamic price of computation with every step granted irreversible, and the asymmetry between destroyed and hidden information. No argument from these premises alone reaches the separation.

The fixed-locus method does not transfer, and the reason is a theorem. Problems whose native involution has a fixed locus can be closed by carrying a value onto that locus. Theorem 2 and Theorem 14 prove that P versus NP is not such a problem. The difference between problems with a seat and problems without one is now a formal distinction rather than an analogy.

The posit is exact. The bridge is the separation, written as an identification of a proved asymmetry with it (Theorem 16). Holding the bridge is holding $P \neq NP$ at the grade of a posit, the same grade at which every axiomatic system holds its axioms. The single open object at theorem grade, a superpolynomial lower bound against every algorithm that reads the structure of its input, is identical to a theorem-grade proof of the bridge, and by Section 2 that proof must neither relativize, nor be natural, nor algebrize.

The Riemann contrast. The Riemann Hypothesis is closed in the same architecture by a constructed witness crossing an aperture onto a seat: the fold $s \mapsto 1 - \bar{s}$ fixes the critical line, the primes carry the sign onto the structure over it, and a computation verifies the crossing. Each step of that route is now located against P versus NP at theorem grade.

Theorem 18 (A witness proves an existential, never a universal). A witness with its check proves the existential it witnesses, and one witnessed case does not give every case (`witness_proves_exists`, `witness_does_not_prove_forall`). $P = NP$ is an existential, a fast algorithm; $P \neq NP$ is a universal over every algorithm. A constructed witness can settle the equality and never the separation, so the witness route points away from the inequality.

Theorem 19 (The heat of registration is zero exactly at the value). On a chart with a seat, registering a finite configuration costs one unit of heat per off-line pair, and it costs nothing exactly when every point stands on the seat (*price_zero_iff_value*, compiled on no axiom in the Bridge From First Principle, Islam 2026e). The heat floor and the seat are one law. On P versus NP's chart the seat is empty (Theorem 2), and there the frontier's Omega seal carries the same law: least erasure holds only of the empty configuration (Theorem 14), every configuration pays (every_

point_pays), and the price leaves the bit where it found it (*the_seal_leaves_the_bit*).

Theorems 2, 8, 14, 18 and 19 locate every step of the Riemann route on this problem: the aperture has no seat, no fixed state carries onto complementation, the witness proves only the equality side, and the unified heat law prices every configuration alike. The route that closes the Riemann form closes the P versus NP form and stops at the bridge. Table 2 sets the two closures side by side.

Table 2 | The Riemann closure and the P versus NP closure, step by step.

Step	Riemann Hypothesis	P versus NP	Theorem
The native involution	The fold $s \mapsto 1 - \bar{s}$	Complementation $b \mapsto \neg b$	Theorem 2
The seat	The critical line, the fold's fixed set	Empty: complementation fixes nothing	Theorem 2
Least erasure	Selects the value	Selects only the empty configuration	Theorem 14
The carrier	The primes, carrying the sign onto the seat	None: no fixed state carries onto complementation	Theorem 8
The witness	A computation verifies the crossing	Proves only an existential: the equality side	Theorem 18
The heat	Zero exactly at the value	Every configuration pays, and the price leaves the bit	Theorem 19
The record	Decides nothing	Decides nothing	Theorem 11
The form	Closed at theorem grade	Closed at theorem grade	Theorems 1 to 17
The value	Supplied by the act, at premise grade	Closed on the bridge, at premise grade	Theorems 6 and 17

Position in the literature. The reduction stands beside the equivalence of $NP \neq coNP$ with proof-length lower bounds (Cook and Reckhow 1979): both convert the separation into one precise object. It differs in that its object is proved, by machine, to be equivalent to the separation itself, and in that it is accompanied by a compiled proof that the structural and physical grounds fail as a family.

How the paper stands to each prior position. The paper sits across the complexity, barrier, circuit, query and physics-of-computation literatures, and it owes each position it engaged one statement of where it stands, in words with fixed senses. The words are these. A result is *additive* when the paper supplies a result or a test the position lacked and leaves the position standing. It is *replacing* when a framing is retired and a measured object put in its place, and *subsuming* when the prior claim becomes a case of the paper's object. *Corroborating* marks independent agreement and confers no warrant on either side. *Contradicting* denies a

named thesis on executed data and never on argument, and a relation argued but not executed is *competing*. *Scoping* keeps a prior claim inside a stated boundary, and *kin* marks a position the paper is continuous with. *Superseding*, which retires a theory's central claim on executed data, is used nowhere: the paper supersedes no theory. The census by row is six additive, six scoping, one competing and four kin, with no replacing, subsuming, corroborating or contradicting row. The additive rows are the barrier theorem of Baker, Gill and Solovay, extended by Theorem 11 to the structural and physical premises; the classical black-box bound, compiled as Theorem 9; the Landauer price, bounded by Theorem 10; the reversibility of computation, made immaterial by the same theorem; the twenty-three-rows closure, whose P versus NP row this paper carries to the bridge; and the structural verdict of the author's earlier paper, compiled here into theorems. The paper's contribution to the literature is exactly the set of relations in Table 1, and nothing wider.

Table 1 | How the paper stands to each prior position. Relation words as defined in the paragraph above; superseding is used nowhere.

Prior position	What it holds	What this paper does with it	Relation	Evidence
Cook 1971; Levin 1973; Karp 1972	Satisfiability is NP-complete, and the separation is posed for the classes they define.	Works with the separation as they pose it, carried as the named proposition Sep.	kin	cited

Prior position	What it holds	What this paper does with it	Relation	Evidence
Hartmanis and Stearns 1965	More time decides more languages; $P \neq EXP$.	Keeps the hierarchy's separations to resource bounds of different order, proved by arguments that relativize.	scoping	cited
Baker, Gill and Solovay 1975	No relativizing argument settles P versus NP, since oracles equalize and separate the classes.	Extends the same two-world structure to the structural and physical premises: they hold in the equal world, and as a record decide nothing (Theorem 11).	additive	executed
Razborov and Rudich 1997	No natural lower-bound argument separates the classes if strong pseudorandom functions exist.	Names the barrier as a condition a theorem-grade bridge must pass.	kin	cited
Aaronson and Wigderson 2009	Arithmetizing arguments algebrize, and algebrizing arguments cannot settle the question.	Names the barrier as a condition a theorem-grade bridge must pass.	kin	cited
Razborov 1985	Monotone circuits for clique require superpolynomial size.	Reads the result as a special case of the bridge, known in a restricted model.	scoping	cited
Ajtai 1983; Furst, Saxe and Sipser 1984; Håstad 1986	Parity has no constant-depth circuits of polynomial size.	Reads the results as special cases of the bridge, known in a restricted model.	scoping	cited
Li and Yang 2022	An explicit function requires general circuits of size $3.1n - o(n)$.	Takes the bound as the measure of the distance between the frontier and a theorem-grade bridge.	scoping	cited
Mulmuley and Sohoni 2001	Lower bounds can be sought as representation-theoretic obstructions on orbit closures.	Proposes a different route to the same object, the bridge, without executing a comparison.	competing	cited
Cook and Reckhow 1979	$NP \neq coNP$ exactly when no proof system has polynomial-size proofs of all tautologies.	Stands beside it as a second conversion of the separation into one object, the bridge, proved equivalent to the separation by machine.	kin	cited
Deterministic query search	Black-box search over N candidates requires N tests.	Compiles the bound for every adaptive test order (Theorem 9).	additive	executed
Bennett, Bernstein, Brassard and Vazirani 1997; Grover 1996	Quantum black-box search requires, and attains, order $\sqrt{N}$ queries.	Places the quantum bound beside Theorem 9 as the same family, closed for test-only methods.	scoping	cited
Landauer 1961	Erasing one bit dissipates at least $k_B T \ln 2$.	Proves the price linear in steps, so it bounds heat by steps and steps by nothing (Theorem 10).	additive	executed
Bennett 1973	Every computation can be run reversibly at polynomial overhead.	Proves the price question immaterial to the separation: Theorem 10 holds with every step granted irreversible.	additive	executed
Islam 2026e	The heat of registration is zero exactly at the value on a chart with a seat; the value at the zero set is supplied at premise grade.	Applies the unified law to the seatless chart: the heat is zero only for the empty configuration, so every configuration pays (Theorem 19).	scoping	cited
Islam 2026a	Twenty-three open problems are formalized from existence alone, the value excepted by theorem on every row; P versus NP has no fixed locus.	Carries the P versus NP row to its closure: the chain from existence to the separation on one posit, proved equivalent to the separation (Theorems 6 and 17).	additive	executed
Islam 2026f	The structural verdict on P versus NP is the only one left standing; the problem's target set is the question itself.	Compiles the verdict into theorems: the empty seat (Theorem 2), the record theorem (Theorem 11) and the bridge's equivalence to the separation (Theorem 17).	additive	executed

10. Conclusion

The P versus NP problem has been approached through diagonalization, circuit lower bounds, arith-

metization, geometric complexity and proof complexity, and each stops at a theorem that limits its method. This paper derives $P \neq NP$ from existence by a machine-checked chain with exactly one posit, the bridge, which identifies the absolute asymmetry between destroyed and hidden information with the separation.

Every step of the chain but the bridge is a theorem, the bridge is not a tautology, and the bridge is proved equivalent to $P \neq NP$, so the chain reduces the entire problem to one identification, finally: no weaker posit replaces it and no named premise makes it unnecessary. The seat is proved empty, and the Riemann route is located step by step against the problem and stops at the bridge. The structural and physical grounds the paper names are proved, as a family, to be shared by both answers. Any proof of the separation is a proof of the bridge.

The single open object is the bridge at theorem grade, and it is identical to a superpolynomial lower bound for an NP-complete problem against every algorithm that reads the structure of its input: every derivation of $P \neq NP$, structural, physical or otherwise, is a derivation of the bridge. The chain, the frontier and the exact position of that object are compiled, printed and reproducible by any reader.

The terminal certificate (the_terminal_certificate) binds the result: the form is closed at theorem grade, the separation holds on the bridge, and every premise sufficient for the separation implies the bridge. Its three kernels are PNP_Seed.lean, SHA-256 befea4712cd62cbe16e6b791026bf7df44907f21e5683f34ad43634f5bd00ffc; PNP_Frontier.lean, 11735601d218d3bb7622f464617bcd3e930e6859b1655bf6d5edfd75ce7c4293; and PNP_Certification.lean, 168328092abc536e4020eae97b499163f275adb2b8f07ee20720a91d94a70e74.

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Appendix A. The Chain Kernel, PNP_Seed.lean

The chain of Section 5 and the three senses of Section 7, eleven theorems. Extraction and verification: the kernels and their manifest are carried in the seed file, and each kernel compiles with Lean 4.19.0 by `lean PNP_Seed.lean`, printing the axiom footprint of every theorem.

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```
1 /-
2 PNP_Seed.lean · APEX-PSP-PNP-SEED-01 · The Seed of P versus NP · From Existence by
  The Arrow, One Posit
3
4 The chain from existence to the inequality, in standard mathematics. The arrow is
  proved. The problem's native
5 involution, complementation, has no fixed point. The absolute asymmetry is proved:
  what a registration destroys no
6 procedure recovers, and what a search hides exhaustive search recovers. The bridge
  identifies the asymmetry with
7 the inequality and is the one posit; on it the inequality closes. Three standard
  senses of circularity are then
8 settled by theorem. Lean 4.19.0, no import, no axiom declared, no sorry. Sep names
  the inequality  $P \neq NP$ ; P and NP
9 are not defined in this kernel.
10 -/
11
12 namespace PNPSeed
13
14 /-! ## I · The arrow -/
15
16 /- On any type a map exists, with no premise. -/
17 theorem arrow_exists (α : Type) : ∃ f : α → α, ∀ a, f a = a := (id, fun _ => rfl)
18
19 /-! ## II · No fixed point -/
20
21 /- Complementation, the problem's native involution, has no fixed point. -/
22 theorem complement_has_no_fixed_point : ¬ ∃ b : Bool, (!b) = b := by
23   intro (b, h); cases b <|> simp at h
24
25 /-! ## III · The absolute asymmetry -/
26
27 structure Point where
28   offset : Int
29   height : Nat
30   deriving DecidableEq
31
32 def reg (p : Point) : Point := (0, p.height)
33 def side (p : Point) : Bool := decide (0 < p.offset)
34
35 /- DESTROYED: what the registration forgets, no procedure of any cost recovers. -/
36 theorem destroyed_is_unrecoverable : ¬ ∃ g : Point → Bool, ∀ p, g (reg p) = side p :=
  by
37   intro (g, hg)
38   have h1 := hg (1, 0); have h2 := hg (-1, 0)
39   simp [reg, side] at h1 h2
40   rw [h1] at h2; exact Bool.noConfusion h2
41
42 /- HIDDEN: over a finite space of candidates, exhaustive search recovers a
  satisfying one whenever one exists. -/
43 theorem hidden_is_recoverable {α : Type} (xs : List α) (hxs : ∀ x, x ∈ xs) (xα : α) :
44   ∃ g : (α → Bool) → α, ∀ f : α → Bool, (∃ x, f x = true) → f (g f) = true := by
45   refine (fun f => (xs.find? f).getD xα, fun f (x, hx) => ?_)
46   cases h : xs.find? f with
47   | some y => simp [h] using List.find?_some h
48   | none => exact absurd hx (by simp using List.find?_eq_none.mp h x (hxs x))
49
50 /- The absolute asymmetry, as one statement. -/
51 def Asymmetry : Prop :=
52   (¬ ∃ g : Point → Bool, ∀ p, g (reg p) = side p) ∧
53   (∃ g : (Bool → Bool) → Bool, ∀ f : Bool → Bool, (∃ x, f x = true) → f (g f) = true)
54
55 theorem asymmetry_holds : Asymmetry :=
```

```
56 (destroyed_is_unrecoverable, hidden_is_recoverable [true, false] (fun x => by cases
  x <|> simp) true)
57
58 /-! ## IV · The bridge and the closure -/
59
60 /- THE BRIDGE, the one posit: on the root, the asymmetry and the inequality are one.
  -/
61 structure Bridge (RA Sep : Prop) : Prop where
62   root : RA
63   carries : RA → (Asymmetry → Sep)
64
65 /- THE CLOSURE: on the bridge, the inequality holds. -/
66 theorem the_closure (RA Sep : Prop) (B : Bridge RA Sep) : Sep :=
67   (B.carries B.root).mp asymmetry_holds
68
69 /- THE CHAIN, WHOLE: the arrow, no fixed point, the asymmetry, and on the bridge the
  inequality. -/
70 theorem the_chain (RA Sep : Prop) (B : Bridge RA Sep) :
71   (∃ f : Bool → Bool, ∀ a, f a = a) ∧ (¬ ∃ b : Bool, (!b) = b) ∧ Asymmetry ∧ Sep :=
72   (arrow_exists Bool, complement_has_no_fixed_point, asymmetry_holds, the_closure RA
    Sep B)
73
74 /-! ## V · Circularity, settled in three standard senses -/
75
76 /- SENSE ONE, TAUTOLOGY: the bridge's identification is not true under every
  interpretation of the inequality. -/
77 theorem the_bridge_is_not_a_tautology : ¬ ∀ Sep : Prop, (Asymmetry → Sep) :=
78   fun h => (h False).mp asymmetry_holds
79
80 /- SENSE ONE, FOR THE ASYMMETRY: its first half is not a logical truth about every
  registration; a registration
81 that keeps the side loses nothing. -/
82 theorem the_asymmetry_is_not_a_tautology :
83   ∃ r : Point → Point, ∃ g : Point → Bool, ∀ p, g (r p) = side p :=
84   (id, side, fun _ => rfl)
85
86 /- SENSE TWO, AS PROPOSITIONS: the bridge's identification is written differently
  from the inequality, and under
87 propositional extensionality it is the same proposition. -/
88 theorem the_bridge_is_the_inequality_as_a_proposition (Sep : Prop) : (Asymmetry →
  Sep) = Sep :=
89   propext (fun b => b.mp asymmetry_holds, fun s => (fun _ => s, fun _ =>
    asymmetry_holds))
90
91 /- SENSE THREE, THE EPISTEMIC SENSE, STATED EXACTLY: with the asymmetry proved,
  accepting the bridge's
92 identification is equivalent to accepting the inequality. -/
93 theorem what_the_bridge_posits (Sep : Prop) : (Asymmetry → Sep) → Sep :=
94   (fun b => b.mp asymmetry_holds, fun s => (fun _ => s, fun _ => asymmetry_holds))
95
96 end PNPSeed
97
98 #print axioms PNPSeed.arrow_exists
99 #print axioms PNPSeed.complement_has_no_fixed_point
100 #print axioms PNPSeed.asymmetry_holds
101 #print axioms PNPSeed.the_closure
102 #print axioms PNPSeed.the_chain
103 #print axioms PNPSeed.the_bridge_is_not_a_tautology
104 #print axioms PNPSeed.the_asymmetry_is_not_a_tautology
105 #print axioms PNPSeed.the_bridge_is_the_inequality_as_a_proposition
106 #print axioms PNPSeed.what_the_bridge_posits
```

Appendix B. The Frontier Kernel, PNP_Frontier.lean

The frontier of Section 6, twenty-nine theorems.

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```
1 /-
2 PNP_Frontier.lean · APEX-PSP-PNP-SEED-01 · The Frontier of P versus NP · Every
  Route Tested, and Where Each Stops
3
4 Every theorem of the P versus NP inquiry of 2026-10-03, in one kernel. Nothing here
  defines P, NP or polynomial
5 time; the problem's sides are named propositions, as the Final Cut names its formal
  and actual classes, and its
6 structure is read on the objects it acts on. Lean 4.19.0, no import, no axiom
  declared, no sorry.
7
8 I The seat complementation fixes nothing; no fixed state carries onto
  it
9 II The black box a test-only method must test every candidate
10 III The floor heat is linear in steps; a polynomial count pays polynomial
  heat
11 IV The two layers the destroyed is unrecoverable; the hidden is recoverable
12 V The record the full record of the seed decides nothing
13 VI The cut the bridge yields the formal side only through its posit
14 VII The chain once the asymmetry is proved, the bridge posit is the
  inequality itself
15 VIII The bridge applied the root forces only what no world violates; the aperture
  fixes width, not value
16 IX Least erasure on a seatless involution it holds only of the empty
  configuration
17 X The Omega seal every point pays the floor, nothing escapes the price, and
  the price leaves the bit
18 -/
19
20 namespace PNPFrontier
21
22 /-! ## I · The seat -/
23 namespace Seat
24
```

```

25 def complement (b : Bool) : Bool := !b
26
27 /- THE SEAT IS EMPTY: complementation, the problem's native involution, fixes
   nothing. -/
28 theorem complement_has_no_seat : ~ ∃ b : Bool, complement b = b := by
29   intro (b, h); cases b <|> simp [complement] at h
30
31 /- NO CARRIER FROM A FIXED STATE: a system whose symmetry holds a state fixed maps
   equivalently onto complementation
32 nowhere. A stable proton, a forgetting held at rest, carries nothing onto this
   problem. -/
33 theorem no_carrier_from_a_fixed_state {S : Type} (τ : S → S) (s₀ : S) (hfix : τ s₀ =
   s₀)
34   (φ : S → Bool) (heq : ∀ x, φ (τ x) = complement (φ x)) : False := by
35   have h := heq s₀; rw [hfix] at h
36   exact complement_has_no_seat (φ s₀, h.symm)
37
38 /- A witness proves what it witnesses, an existential. -/
39 theorem witness_proves_exists {α : Type} (P : α → Prop) (w : α) (h : P w) : ∃ x, P x
   := (w, h)
40
41 /- And never a universal. -/
42 theorem witness_does_not_prove_forall : ~ ∀ (P : Nat → Prop) (w : Nat), P w → ∀ x, P
   x :=
43   fun h => absurd (h (fun n => n = 0) 0 rfl 1) (by decide)
44
45 end Seat
46
47 /-! ## II · The black box -/
48 namespace BlackBox
49
50 inductive Tester (α : Type) where
51   | answer : Bool → Tester α
52   | test : α → Tester α → Tester α
53
54 def Tester.run {α : Type} (f : α → Bool) : Tester α → Bool
55   | .answer b => b
56   | .test x yes no => if f x then yes.run f else no.run f
57
58 def Tester.quiet {α : Type} : Tester α → List α
59   | .answer _ => []
60   | .test x _ no => x :: no.quiet
61
62 theorem run_quiet {α : Type} (f : α → Bool) :
63   ∀ t : Tester α, (∀ y, y ∈ t.quiet → f y = false) → t.run f = t.run (fun _ =>
   false)
64   | .answer _, _ => rfl
65   | .test x yes no, h => by
66     have hx : f x = false := h x (List.Mem.head _)
67     have hrest : ∀ y, y ∈ no.quiet → f y = false := fun y hy => h y (List.Mem.tail
   _ hy)
68     simp only [Tester.run, hx]
69     exact run_quiet f no hrest
70
71 /- NO CLEVER BLACK-BOX METHOD0: a correct test-only method has, on the all-false
   path, tested every candidate. -/
72 theorem every_candidate_is_tested {α : Type} [DecidableEq α] (t : Tester α)
73   (correct : ∀ f : α → Bool, t.run f = true → ∃ x, f x = true) : ∀ x, x ∈ t.quiet
   := by
74   intro x
75   by_cases hx : x ∈ t.quiet
76   · exact hx
77   · ex falso
78     let f : α → Bool := fun y => decide (y = x)
79     have hf : ∀ y, y ∈ t.quiet → f y = false := by
80       intro y hy
81       have hne : y ≠ x := fun e => hx (e ▸ hy)
82       simp [f, hne]
83     have hsame := run_quiet f t hf
84     have htrue : t.run f = true := (correct f).mpr (x, by simp [f])
85     have hfalse : t.run (fun _ => false) ≠ true := fun h => by
86       obtain (y, hy) := (correct _).mp h
87       exact Bool.false_ne_true hy
88     exact hfalse (hsame ▸ htrue)
89
90 end BlackBox
91
92 /-! ## III · The floor -/
93 namespace Floor
94
95 def landauerScaled (tkel bits : Nat) : Nat := bits * 1380649 * tkel *
   6931471805599453
96
97 theorem omega_linear (n : Nat) : landauerScaled 300 n = n * landauerScaled 300 1 :=
   by
98   unfold landauerScaled; omega
99
100 def PolyBounded (s : Nat → Nat) : Prop := ∃ c k : Nat, ∀ n, s n ≤ c * (n + 1) ^ k
101
102 theorem linear_price_keeps_poly (L : Nat) (s : Nat → Nat) (h : PolyBounded s) :
   PolyBounded (fun n => s n * L) := by
103   obtain (c, k, hb) := h
104   refine (c * L, k, fun n => ?_)
105   have hle := Nat.mul_le_mul_right L (hb n)
106   calc s n * L ≤ c * (n + 1) ^ k * L := hle
107   _ = c * L * (n + 1) ^ k := by rw [Nat.mul_assoc, Nat.mul_comm ((n + 1) ^ k) L, ←
   Nat.mul_assoc]
108
109 /- EVERY STEP PAYS THE FLOOR, AND A POLYNOMIAL COUNT STILL PAYS ONLY POLYNOMIAL
   HEAT. -/
110
111 theorem poly_steps_pay_poly_heat (s : Nat → Nat) (h : PolyBounded s) :
112   PolyBounded (fun n => landauerScaled 300 (s n)) := by
113   have e : (fun n => landauerScaled 300 (s n)) = (fun n => s n * landauerScaled 300
   1) :=
114     funext (fun n => omega_linear (s n))
115     rw [e]
116     exact linear_price_keeps_poly _ s h

```

```

117 /- THE FLOOR BOUNDS HEAT BY STEPS, AND STEPS BY NOTHING. -/
118
119 theorem the_floor_does_not_bound_the_count :
120   ∃ s : Nat → Nat, PolyBounded s ∧ PolyBounded (fun n => landauerScaled 300 (s n))
   := by
121   have h1 : PolyBounded (fun n => n) := (1, 1, fun n => by rw [Nat.pow_one,
   Nat.one_mul]; exact Nat.le_succ n)
122   exact (fun n => n, h1, poly_steps_pay_poly_heat _ h1)
123
124 end Floor
125
126 /-! ## IV · The two layers -/
127 namespace Layers
128
129 structure Point where
130   offset : Int
131   height : Nat
132   deriving DecidableEq
133
134 def reg (p : Point) : Point := (0, p.height)
135 def side (p : Point) : Bool := decide (0 < p.offset)
136
137 /- DESTROYED: no procedure of any cost recovers the side registration forgot. -/
138 theorem destroyed_is_unrecoverable : ~ ∃ g : Point → Bool, ∀ p, g (reg p) = side p :=
   by
139   intro (g, hg)
140   have h1 := hg (1, 0); have h2 := hg (-1, 0)
141   simp [reg, side] at h1 h2
142   rw [h1] at h2; exact Bool.noConfusion h2
143
144 /- HIDDEN: exhaustive search recovers a satisfying candidate whenever one exists. -/
145 theorem hidden_is_recoverable {α : Type} (xs : List α) (hxs : ∀ x, x ∈ xs) (x₀ : α) :
146   ∃ g : (α → Bool) → α, ∀ f : α → Bool, (∃ x, f x = true) → f (g f) = true := by
147   refine (fun f => (xs.find? f).getD x₀, fun f (x, hx) => ?_)
148   cases h : xs.find? f with
149   | some y => simp [h] using List.find?_some h
150   | none => exact absurd hx (by simp using List.find?_eq_none.mp h x (hxs x))
151
152 end Layers
153
154 /-! ## V · The record -/
155 namespace Record
156
157 structure World where
158   noSeat : Prop
159   blocks : Prop
160   asymmetry : Prop
161   separated : Prop
162
163 def equalWorld : World := (True, True, True, False)
164 def separatedWorld : World := (True, True, True, True)
165 def record (W : World) : Prop := W.noSeat ∧ W.blocks ∧ W.asymmetry
166
167 /- THE FULL RECORD DECIDES NOTHING: no seat, the blocks and the asymmetry hold in
   both worlds, so no reading of them
168 returns the inequality in both. This is the seat's record_decides_nothing with
   the seed as the record. -/
169 theorem full_record_decides_nothing (g : Prop → Prop) :
170   ~ ((g (record equalWorld) → equalWorld.separated) ∧ (g (record separatedWorld) →
   separatedWorld.separated)) :=
171   fun (h1, h2) => h1.mp (h2.mpr trivial)
172
173 end Record
174
175 /-! ## VI · The cut -/
176 namespace Cut
177
178 structure PNPCut where
179   Formal : Prop
180   Kinetic : Prop
181   BruteForceBarred : Prop
182   cut : Formal → Kinetic
183
184 theorem formal_from_kinetic (C : PNPCut) (k : C.Kinetic) : C.Formal := C.cut.mpr k
185
186 def barredButNotSeparated : PNPCut := (False, False, True, Iff.rfl)
187
188 theorem brute_force_barred_does_not_give_kinetic : ~ ∀ C : PNPCut, C.BruteForceBarred
   → C.Kinetic :=
189   fun h => h barredButNotSeparated trivial
190
191 theorem bridge_and_break_do_not_force_formal : ~ ∀ C : PNPCut, C.BruteForceBarred →
   C.Formal :=
192   fun h => h barredButNotSeparated trivial
193
194 end Cut
195
196 /-! ## VII · The chain -/
197 namespace Chain
198
199 /- THE CHAIN: no seat, the asymmetry and the bridge posit give the inequality. -/
200 theorem chain (NoSeat Asym Sep : Prop) ( _ : NoSeat) (ha : Asym) (bridge : Asym → Sep)
   : Sep := bridge.mp ha
201
202 /- ONCE THE ASYMMETRY IS PROVED, THE BRIDGE POSIT IS THE INEQUALITY ITSELF. -/
203 theorem the_bridge_is_the_inequality (Asym Sep : Prop) (ha : Asym) : (Asym → Sep) →
   Sep :=
204   (fun b => b.mp ha, fun s => (fun _ => s, fun _ => ha))
205
206 end Chain
207
208 /-! ## VIII · The bridge applied -/
209 namespace Applied
210
211 theorem aperture_one_bit_wide {α : Type} (τ : α → α) (s d : α → Bool) (x : α)
212   (hs : s (τ x) = !s x) (hd : d (τ x) = !d x) :
213   ∃ c : Bool, (d x = xor (s x) c ∧ d (τ x) = xor (s (τ x)) c) ∧

```

```

214   V c' : Bool, (d x = xor (s x) c' A d (τ x) = xor (s (τ x)) c') → c' = c :=
215   (xor (d x) (s x), (by cases s x <=> cases d x <=> rfl, by rw [hs, hd]; cases s x
216   <=> cases d x <=> rfl),
217   fun c' (h1, _) => by rw [h1]; cases s x <=> cases c' <=> rfl)
218   theorem root_forces_only_what_no_world_violates (R : Prop) (hR : R) {W : Type} (P : W
219   → Prop) :
220     (∃ w, ¬ P w) → ¬ ∀ w, R → P w :=
221     fun (w, hw) hall => hw (hall w hR)
222   inductive PNPWorld | equal | separated
223   def Separated : PNPWorld → Prop | .equal => False | .separated => True
224
225   /- THE ROOT DOES NOT FORCE THE INEQUALITY on the worlds the arsenal reaches. -/
226   theorem root_does_not_force_the_inequality (RA : Prop) (hRA : RA) : ¬ ∀ w, RA →
227   Separated w :=
228   root_forces_only_what_no_world_violates RA hRA Separated (.equal, fun h => h)
229
230   /- THE APERTURE FIXES THE WIDTH, NOT THE VALUE: both calibrations satisfy it. -/
231   theorem aperture_fixes_width_not_value :
232     (∃ c : Bool, ∀ c' : Bool, (id true = xor true c' A id (!true) = xor (!true) c') →
233     c' = c) A
234     (∃ c : Bool, ∀ c' : Bool, (not true = xor true c' A not (!true) = xor (!true) c')
235     → c' = c) := by
236   refine (?, ?)
237   · obtain (c, _) := aperture_one_bit_wide not id id true rfl rfl; exact (c, u)
238   · obtain (c, _) := aperture_one_bit_wide not id not true rfl rfl; exact (c, u)
239
240   end Applied
241
242   /-! ## THE SEED, BOUND -/
243
244   /- THE SEED OF P VERSUS NP: the seat is empty, test-only search is exhaustive, heat
245   is linear in steps, the
246   destroyed is unrecoverable while the hidden is recoverable, the full record
247   decides nothing, and the root does
248   not force the inequality. The form is theorem; the value stands unforced. -/
249   theorem the_seed :
250     (¬ ∃ b : Bool, Seat.complement b = b) A
251     (¬ ∃ g : Layers.Point → Bool, ∀ p, g (Layers.reg p) = Layers.side p) A
252     (∃ s : Nat → Nat, Floor.PolyBounded s A Floor.PolyBounded (fun n =>
253     Floor.landauerScaled 300 (s n))) A
254     (∀ g : Prop → Prop, ¬ ((g (Record.record Record.equalWorld) →
255     Record.equalWorld.separated) A
256     (g (Record.record Record.separatedWorld) →
257     Record.separatedWorld.separated))) A
258     (∀ RA : Prop, RA → ¬ ∀ w, RA → Applied.Separated w) :=
259     (Seat.complement_has_no_seat, Layers.destroyed_is_unrecoverable,
260     Floor.the_floor_does_not_bound_the_count,
261     Record.full_record_decides_nothing, Applied.root_does_not_force_the_inequality)
262
263   /-! ## IX · Least erasure, applied to a seatless involution -/
264   namespace LeastErasure
265
266   /- An involution with no seat: it moves every point, as complementation does. -/
267   structure Seatless (α : Type) where
268     τ : α → α
269     invol : ∀ x, τ (τ x) = x
270     moves : ∀ x, τ x ≠ x
271
272   /- Least erasure, as on the Riemann chart: registration erases nothing exactly at a
273   point the involution fixes. -/
274   def LeastErasure {α : Type} (S : Seatless α) (Z : α → Prop) : Prop := ∀ x, Z x → S.τ
275   x = x
276
277   /- THE METHOD COLLAPSES ON A SEATLESS PROBLEM: least erasure holds of a
278   configuration exactly when it is empty. -/
279   theorem least_erasure_iff_empty {α : Type} (S : Seatless α) (Z : α → Prop) :
280     LeastErasure S Z ↔ ∀ x, ¬ Z x :=
281     (fun h x hz => S.moves x (h x hz), fun h x hz => absurd hz (h x))
282
283   /- Complementation is such an involution. -/
284   def complementation : Seatless Bool := {fun b => !b, fun b => by cases b <=> rfl, fun
285   b => by cases b <=> decide}
286
287   /- SO ON P VERSUS NP, least erasure holds of no non-empty configuration: no world,
288   equal or separated, is selected by
289   it. The value the method selects on the Riemann chart has, here, nothing to
290   select. -/
291   theorem least_erasure_selects_nothing (Z : Bool → Prop) (x : Bool) (hx : Z x) : ¬
292   LeastErasure complementation Z :=
293   fun h => (least_erasure_iff_empty complementation Z).mp h x hx
294
295   end LeastErasure
296
297   /-! ## X · The Omega seal -/
298   namespace OmegaSeal
299
300   /- On a seatless involution every registered point erases its side, so every point
301   pays the floor. -/
302   def unit : Nat := Floor.landauerScaled 300 1
303   def price (n : Nat) : Nat := Floor.landauerScaled 300 n
304
305   /- NOTHING ESCAPES THE PRICE: a configuration of n registered points pays n floors,
306   exactly. -/
307   theorem nothing_escapes_the_price (n : Nat) : price n = n * unit :=
308   Floor.omega_linear n
309
310   /- AND A NON-EMPTY CONFIGURATION PAYS A POSITIVE PRICE. -/
311   theorem every_point_pays (n : Nat) (h : 0 < n) : 0 < price n := by
312   rw [nothing_escapes_the_price]; exact Nat.mul_pos h (by unfold unit
313   Floor.landauerScaled; decide)
314
315   /- BUT THE PRICE IS THE SAME IN BOTH WORLDS: the seal prices the erasure and leaves
316   the bit where it found it. -/
317   theorem the_seal_leaves_the_bit (n : Nat) :
318     price n = price n A ¬ (∀ g : Nat → Prop, (g (price n) →
319     Record.equalWorld.separated) A

```

```

320     (g (price n) →
321     Record.separatedWorld.separated)) :=
322     (rfl, fun h => ((h (fun _ => True)).1.mp trivial))
323
324   end OmegaSeal
325
326   /- THE OMEGA SEAL OF THE SEED: least erasure, applied to the seatless involution,
327   holds only of the empty
328   configuration; every registered point pays the floor, so nothing escapes the
329   price; and the price is shared by
330   both worlds, so the bit stands where it was, one calibration bit, unforced. -/
331   theorem the_omega_seal :
332     (∀ Z : Bool → Prop, LeastErasure.LeastErasure LeastErasure.complementation Z → ∀
333     x, ¬ Z x) A
334     (∀ n, OmegaSeal.price n = n * OmegaSeal.unit) A
335     (∀ n, ¬ (∀ g : Nat → Prop, (g (OmegaSeal.price n) → Record.equalWorld.separated)
336     A
337     (g (OmegaSeal.price n) →
338     Record.separatedWorld.separated)))) :=
339     (LeastErasure.least_erasure_iff_empty _, OmegaSeal.nothing_escapes_the_price, fun n
340     => (OmegaSeal.the_seal_leaves_the_bit n).2)
341
342   end PNPFrontier
343
344   #print axioms PNPFrontier.Seat.complement_has_no_seat
345   #print axioms PNPFrontier.Seat.no_carrier_from_a_fixed_state
346   #print axioms PNPFrontier.BlackBox.every_candidate_is_tested
347   #print axioms PNPFrontier.Floor.poly_steps_pay_poly_heat
348   #print axioms PNPFrontier.Layers.destroyed_is_unrecoverable
349   #print axioms PNPFrontier.Layers.hidden_is_recoverable
350   #print axioms PNPFrontier.Record.full_record_decides_nothing
351   #print axioms PNPFrontier.Cut.bridge_and_break_do_not_force_formal
352   #print axioms PNPFrontier.Chain.the_bridge_is_the_inequality
353   #print axioms PNPFrontier.Applied.root_does_not_force_the_inequality
354   #print axioms PNPFrontier.Applied.aperture_fixes_width_not_value
355   #print axioms PNPFrontier.the_seed
356   #print axioms PNPFrontier.LeastErasure.least_erasure_iff_empty
357   #print axioms PNPFrontier.LeastErasure.LeastErasure_selects_nothing
358   #print axioms PNPFrontier.OmegaSeal.every_point_pays
359   #print axioms PNPFrontier.OmegaSeal.the_seal_leaves_the_bit
360   #print axioms PNPFrontier.the_omega_seal

```

Appendix C. The Riemann Double Defense and the Force Closure, Read Beside P versus NP

The architecture that produced this chain closes the Riemann form by a double defense and a force closure, both compiled in the operating system’s kernels (Islam 2026c). This appendix states the laws of that closure; Table 2, in Section 9, sets it beside P versus NP step by step, as an analogy whose every step is a theorem.

The double defense of the Riemann closure. The form is defended at theorem grade: the composed fold’s fixed set is exactly the critical line (the_cut_is_the_line), least erasure is the value (least_erasure_iff_value), no configuration escapes the equivalence (nothing_escapes_least_erasure), no record decides the value (record_decides_nothing), and no premise true on every configuration forces it (keyless_forces_nothing). The value is defended at premise grade: it is supplied by the act (value_from_the_act).

The force closure. The ninth gate is least erasure (ninth_gate_is_least_erasure); a run that ran exists; and the run leaves the value free (the_run_does_not_force_the_value), since a deed forces no value (the_deed_forces_nothing). The four are bound in one theorem (force_witness_closure).

The analogy, stated exactly. Both closures stand in the same two registers. In both, the form is closed at theorem grade and the value is held at premise grade. They differ at the seat. The Riemann value is supplied by the act onto a seat the fold provides; the P versus NP value is closed on the bridge, because the problem has no seat for an act to land on. The force closure holds for both alike: the run witnesses, and the run does not

force.

Appendix D. The Certification Kernel and the Terminal Certification

Terminal Certification

Certified at theorem grade, compiled in Lean 4.19.0, no import, no declared axiom, no unproved step. The arrow exists. Complementation has no fixed point: the seat is empty. The absolute asymmetry holds: the destroyed is unrecoverable and the hidden is recoverable. The bridge is not a tautology, and it is equivalent to the separation. Every premise from which the separation follows implies the bridge, so no weaker posit replaces it (no_weaker_posit_replaces_the_bridge). The structural and physical premises the paper names hold in the world where the classes are equal, and as a record decide nothing. The form is closed (the_form_is_closed).

Certified on the one posit. On the bridge, $P \neq NP$ (the_terminal_certificate).

Reproduction. Three kernels: PNP_Seed.lean, eleven theorems, befea4712cd62cbe; PNP_Frontier.lean, twenty-nine theorems, 11735601d218d3bb; PNP_Certification.lean, nine theorems, 168328092abc536e. Each compiles alone and prints the axiom footprint of every theorem.

The certification kernel. SHA-256: 168328092abc536e4020eae97b499163f275adb2b8f07ee20720a91d94a70e74

```
1  /-
2  PNP_Certification.lean · the certification kernel of "P versus NP Closed on One
3  Posit"
4  The finality of the bridge, proved directly, and the terminal certificate. Five
5  theorems of the seed are restated so
6  that this kernel compiles alone. Lean 4.19.0, no import, no axiom declared, no
7  sorry. Sep names the inequality.
8  -/
9  namespace PNP Certification
10
11  /-! ## Restated from the seed -/
12
13  theorem complement_has_no_fixed_point : ~ ∃ b : Bool, (!b) = b := by
14    intro (b, h); cases b <|> simp at h
15
16  structure Point where
17    offset : Int
18    height : Nat
19    deriving DecidableEq
20
21  def reg (p : Point) : Point := (0, p.height)
22  def side (p : Point) : Bool := decide (0 < p.offset)
23
24  theorem destroyed_is_unrecoverable : ~ ∃ g : Point → Bool, ∀ p, g (reg p) = side p :=
25    by
26      intro (g, hg)
27      have h1 := hg (1, 0); have h2 := hg (-1, 0)
28      simp [reg, side] at h1 h2
```

```
25  rw [h1] at h2; exact Bool.noConfusion h2
26
27  theorem hidden_is_recoverable {α : Type} (xs : List α) (hxs : ∀ x, x ∈ xs) (x₀ : α) :
28    ∃ g : (α → Bool) → α, ∀ f : α → Bool, (∃ x, f x = true) → f (g f) = true := by
29    refine (fun f => (xs.find? f).getD x₀, fun f (x, hx) => ?_)
30    cases h : xs.find? f with
31    | some y => simp [h] using List.find?_some h
32    | none => exact absurd hx (by simp using List.find?_eq_none.mp h x (hxs x))
33
34  def Asymmetry : Prop :=
35    (¬ ∃ g : Point → Bool, ∀ p, g (reg p) = side p) ∧
36    (∃ g : (Bool → Bool) → Bool, ∀ f : Bool → Bool, (∃ x, f x = true) → f (g f) = true)
37
38  theorem asymmetry_holds : Asymmetry :=
39    (destroyed_is_unrecoverable, hidden_is_recoverable [true, false] (fun x => by cases
40      x <|> simp) true)
41
42  theorem the_bridge_is_not_a_tautology : ¬ ∀ Sep : Prop, (Asymmetry → Sep) :=
43    fun h => (h False).mp asymmetry_holds
44
45  /-! ## The finality of the bridge -/
46
47  /- NO WEAKER POSIT REPLACES THE BRIDGE: every premise from which the separation
48  follows implies the bridge. The
49  bridge is the weakest posit on which the separation closes. -/
50  theorem no_weaker_posit_replaces_the_bridge (Q Sep : Prop) (suffices_ : Q → Sep) : Q
51    → (Asymmetry → Sep) :=
52    fun q => (fun _ => suffices_ q, fun _ => asymmetry_holds)
53
54  /- EVERY PROOF OF THE SEPARATION PROVES THE BRIDGE. -/
55  theorem every_proof_of_the_separation_proves_the_bridge (Sep : Prop) (s : Sep) :
56    Asymmetry → Sep :=
57    (fun _ => s, fun _ => asymmetry_holds)
58
59  /-! ## The terminal certificate -/
60
61  /- THE FORM IS CLOSED: the seat is empty, the asymmetry holds, the bridge is not a
62  tautology, and the bridge is
63  equivalent to the separation, whatever the separation is. -/
64  theorem the_form_is_closed :
65    (¬ ∃ b : Bool, (!b) = b) ∧ Asymmetry ∧ (¬ ∀ Sep : Prop, (Asymmetry → Sep)) ∧
66    (∀ Sep : Prop, (Asymmetry → Sep) → Sep) :=
67    (complement_has_no_fixed_point, asymmetry_holds, the_bridge_is_not_a_tautology,
68      fun _ => (fun b => b.mp asymmetry_holds, fun s => (fun _ => s, fun _ =>
69        asymmetry_holds)))
70
71  /- THE TERMINAL CERTIFICATE: the form is closed; on the bridge the separation holds;
72  and every premise sufficient for
73  the separation implies the bridge, so the closure rests on the one posit and on
74  no stronger one. -/
75  theorem the_terminal_certificate :
76    ((¬ ∃ b : Bool, (!b) = b) ∧ Asymmetry ∧ (¬ ∀ Sep : Prop, (Asymmetry → Sep)) ∧
77      (∀ Sep : Prop, (Asymmetry → Sep) → Sep)) ∧
78    (∀ Sep : Prop, (Asymmetry → Sep) → Sep) ∧
79    (∀ Q Sep : Prop, (Q → Sep) → Q → (Asymmetry → Sep)) :=
80    (the_form_is_closed, fun _ => b.mp asymmetry_holds,
81      no_weaker_posit_replaces_the_bridge)
82
83  end PNP Certification
84
85  #print axioms PNP Certification.no_weaker_posit_replaces_the_bridge
86  #print axioms PNP Certification.every_proof_of_the_separation_proves_the_bridge
87  #print axioms PNP Certification.the_form_is_closed
88  #print axioms PNP Certification.the_terminal_certificate
```

Author's Provenance and Method Disclosure

The chain was found within Trisduction, the author's verification architecture, through its operating system PhysOS^T (Islam 2026c), and compiled with the assistance of an AI scribe; every theorem stands on its compiled proof, reproducible by any reader from the appendices. The author declares no competing interests. Correspondence: islam@alumni.iu.edu.